Project 1 The Evolution of Image Classification

From Handcrafted Features to Convolutional Neural Networks

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Complete Project Report

1 Introduction

This project studies how image representation and model architecture affect classification as visual complexity increases. I compared raw pixels, Histogram of Oriented Gradients (HOG), and learned neural-network representations using MNIST and CIFAR-10. MNIST contains centered grayscale digits, while CIFAR-10 contains small color photographs with varied objects, backgrounds, viewpoints, and variation within each class.

The completed experiments cover five classification pipelines on each dataset: raw-pixel logistic regression, HOG with logistic regression, HOG with a linear support vector machine, a fully connected neural network, and a convolutional neural network. Changing the representation from raw pixels to HOG produced a much larger improvement than changing between the two linear classifiers. The CNN achieved the highest test accuracy on both datasets, with its largest advantage on CIFAR-10. Overall, the results show that representing local and spatial image structure becomes increasingly important as visual complexity increases.

2 Experimental Protocol

The official torchvision MNIST and CIFAR-10 datasets were used. Each original training set was divided into an 80% training subset and a 20% validation subset with a stratified split and random seed 1234. This produced 48,000 training and 12,000 validation images for MNIST, and 40,000 training and 10,000 validation images for CIFAR-10. The official 10,000-image test set for each dataset was kept separate until a final configuration had been selected using validation accuracy.

The same fixed splits were reused throughout the project. Classical models were implemented with scikit-learn. StandardScaler was fitted only on training features and then applied to validation and test features through a pipeline. Neural networks were implemented in PyTorch and trained with mini-batches. The final neural-network experiments used a Colab T4 GPU, while the scikit-learn and HOG computations continued to run on the CPU.

3 Work Package 1 Raw Pixel Linear Baselines

3.1 Method

MNIST images are 28 by 28 pixels with one grayscale value per pixel, so flattening each image produced 28 x 28 = 784 features. CIFAR-10 images are 32 by 32 pixels with three color channels (red, green, and blue), so flattening each image produced 32 x 32 x 3 = 3,072 features. Each flattened image became one row of input for the classifier.

StandardScaler was fitted to the training data and standardized each pixel feature by subtracting its training-set mean and dividing by its standard deviation. This centered the features around zero and placed them on more comparable scales, which helped the logistic-regression optimizer adjust the weights and converge more reliably. The same fitted scaler was then applied to the validation and test data. Multiclass logistic regression was trained with the default C value of 1.0, max_iter set to 1,000, and tolerance set to 0.001.

For each image, multiclass logistic regression calculated one score for each of the ten possible classes. Each score was produced by multiplying every standardized feature value by its learned weight for that class, summing those products, and adding a bias. During training, softmax converted the ten scores into probabilities, and the loss measured how much probability the model assigned to the correct label. The optimizer repeatedly adjusted all weights and biases to reduce this loss. Training stopped when the optimizer met the convergence tolerance of 0.001, or when it reached the maximum of 1,000 iterations. For prediction, the class with the highest score was selected. Because each score is a weighted linear combination of the input features, the model learns linear decision boundaries between classes.

3.2 Results

Dataset	Features	Validation	Test	Training time
MNIST	784	91.36%	92.30%	10.17 s
CIFAR-10	3,072	36.24%	36.55%	304.80 s

Table 1. Raw-pixel logistic-regression results

Source: project notebook outputs.

3.3 Analysis

Raw pixels provided a useful baseline but exposed the limits of a linear model. MNIST reached 92.30% test accuracy because the digits are centered within each image and have simple, consistent shapes. A pixel at a particular location therefore has a relatively stable meaning across many MNIST images. MNIST uses one grayscale brightness value per pixel, so each pixel location contributes only one feature indicating how light or dark that position is. This simpler representation allows the classifier to focus on the digit's shape and brightness pattern without also having to account for differences among color channels.

CIFAR-10 reached only 36.55% test accuracy. Objects may appear in different locations and against different backgrounds, so the same pixel location can represent unrelated content across images. Flattening preserves all pixel values, but it removes their explicit two-dimensional arrangement. Logistic regression therefore treats each pixel position as an independent feature without directly modeling relationships between neighboring pixels. A linear classifier cannot directly model nonlinear combinations of shapes, textures, and object parts from those raw pixel positions. CIFAR-10 also uses three color channels per pixel, which increases the number of features and the amount of variation the model must handle. The much longer CIFAR-10 fitting time reflects these larger feature vectors and more difficult decision boundaries.

Overall, raw-pixel logistic regression worked well when the images were simple and consistently positioned, as in MNIST. Its performance dropped sharply on CIFAR-10 because object position, background, color, and appearance varied much more between images. These results show that raw pixels can produce useful linear decision boundaries for highly structured datasets, but they are less effective when classification depends on more complex visual patterns.

4 Work Package 2 HOG Representations

4.1 Method

HOG replaced the raw pixel values with handcrafted features that describe local edge direction and strength. For each pixel, HOG estimated the horizontal and vertical change in brightness. It combined those changes into a gradient magnitude, which describes how strong the brightness change is, and a gradient orientation, which describes its direction. CIFAR-10 images were converted from RGB to grayscale before this step so that each pixel had one brightness value.

Each image was divided into cells containing 4 by 4 pixels. Within each cell, every pixel gradient voted for an orientation bin, with its magnitude determining the strength of the vote. Strong edges therefore had more influence than small brightness changes. The resulting histogram summarized the edge directions found in that local region.

Neighboring cells were grouped into overlapping blocks of 2 by 2 cells. L2-Hys block normalization scaled the combined cell histograms, clipped unusually large values, and scaled them again. This reduced sensitivity to lighting and contrast and kept one very strong gradient from dominating a block. The normalized values from all blocks were then concatenated into one HOG feature vector for each image.

Orientation-bin counts of 6, 9, and 12 were compared separately for each dataset. More bins can describe edge directions more precisely, but they also produce longer feature vectors and give the classifier more weights to learn. Because CIFAR-10 images are larger than MNIST images, they contain more cells and blocks and produce longer HOG vectors for the same number of bins. Pixels per cell, cells per block, block normalization, logistic-regression settings, and the fixed data splits were held constant.

For each candidate, StandardScaler was fitted to the training HOG features and then applied to the validation and test features through the same logistic-regression pipeline used in Work Package 1. Each model was fitted on the training subset, and validation accuracy selected the orientation count independently for MNIST and CIFAR-10. The test set was used only after that selection, preventing test performance from influencing the chosen configuration.

4.2 Candidate Results

Dataset	Bins	Vector length	Train	Validation	Training time
MNIST	6	864	98.53%	97.41%	3.84 s
MNIST	9	1,296	98.80%	97.66%	5.33 s
MNIST	12	1,728	99.10%	97.71%	7.08 s
CIFAR-10	6	1,176	59.79%	51.57%	29.04 s
CIFAR-10	9	1,764	64.40%	51.40%	49.53 s
CIFAR-10	12	2,352	67.68%	50.66%	60.48 s

Table 2. HOG orientation-bin candidates

Source: project notebook outputs.

Dataset	Selected bins	Vector length	Validation	Final test	Training time
MNIST	12	1,728	97.71%	97.86%	7.08 s
CIFAR-10	6	1,176	51.57%	50.84%	29.04 s

Table 3. Selected HOG configurations and required evidence

Source: project notebook outputs.

Increasing the orientation count increased the feature-vector length for both datasets. On MNIST, validation accuracy rose slightly from 97.41% with 6 bins to 97.71% with 12 bins, so 12 bins were selected. On CIFAR-10, 6 bins produced the highest validation accuracy, 51.57%. Although CIFAR-10 training accuracy continued to rise as more bins were added, validation accuracy declined, so the additional directional detail was not used to select the model.

After the orientation counts were selected, the MNIST pipeline reached 97.86% test accuracy and the CIFAR-10 pipeline reached 50.84%. Both test results were close to the validation accuracies used for selection, indicating that the selected configurations behaved similarly on unseen test images.

4.3 Comparison with Raw Pixels

Dataset	Raw-pixel test	Selected HOG test	Accuracy change	Raw-pixel time	HOG time
MNIST	92.30%	97.86%	+5.56 points	10.17 s	7.08 s
CIFAR-10	36.55%	50.84%	+14.29 points	304.80 s	29.04 s

Table 4. Raw-pixel and selected HOG logistic-regression comparison

Source: project notebook outputs.

Keeping logistic regression fixed, HOG increased test accuracy from 92.30% to 97.86% on MNIST and from 36.55% to 50.84% on CIFAR-10. Recorded training time also decreased from 10.17 to 7.08 seconds on MNIST and from 304.80 to 29.04 seconds on CIFAR-10. This comparison isolates the effect of changing the image representation.

4.4 Analysis

The 5.56-point improvement on MNIST was meaningful but smaller than the 14.29-point improvement on CIFAR-10. MNIST digits are already centered and consistently shaped, so their raw pixel positions contain useful information. HOG still improved the result by emphasizing digit edges and shapes instead of relying only on exact brightness values.

CIFAR-10 objects vary more in position, background, lighting, and appearance, making a fixed raw pixel location less reliable. HOG summarizes edge patterns within local regions, so small shifts and brightness differences have less effect on the feature values. The selected CIFAR-10 HOG representation also reduced the feature vector from 3,072 raw values to 1,176 values. Logistic regression remained a linear classifier, but HOG gave it features that were easier to separate with linear decision boundaries.

Increasing the number of orientation bins did not consistently improve validation performance. MNIST benefited slightly from the added directional detail, while the CIFAR-10 models fitted the training data more closely without improving on validation images. HOG preserves local edge direction, edge strength, and rough spatial organization through its cells and blocks. It discards exact pixel brightness, some fine image detail, and, after grayscale conversion, CIFAR-10 color information. These trade-offs explain both its improvement over raw pixels and its remaining limitations on complex natural images.

5 Work Package 3 Selected HOG with Linear SVM

5.1 Method

The selected HOG representation was kept fixed while logistic regression was replaced with a linear support vector machine implemented with LinearSVC. MNIST therefore reused 12 orientation bins and a 1,728-feature vector, while CIFAR-10 reused 6 bins and a 1,176-feature vector. StandardScaler remained in the pipeline so that the HOG features were placed on comparable scales before SVM training.

For each class, LinearSVC learned a weighted equation and bias that defined a decision boundary. Training attempted to create a wide margin while reducing margin violations. For the current boundary, correctly classified examples beyond the margin produced no margin loss, while examples on or inside the margin, including misclassified examples, produced loss and affected the weight updates. These influential examples near or within the margin played the support-vector role. For prediction, LinearSVC calculated a decision score for each of the ten one-versus-rest class boundaries and selected the class with the highest score.

The C value determined how strongly margin violations were penalized compared with the goal of keeping the weights small and the margin wide. A smaller C applied stronger regularization and allowed more violations in exchange for a wider, simpler boundary. A larger C gave violations more influence, encouraging the model to fit the training data more closely but potentially increasing overfitting and training difficulty. Because C changed the learned weights and boundary, it could also change which examples ultimately fell inside or outside the margin. C values of 0.1, 1, and 10 were compared with max_iter set to 10,000, tolerance set to 0.001, and dual set to False.

Each candidate was fitted on the training subset, and both training and validation accuracy were recorded. Validation accuracy selected C independently for MNIST and CIFAR-10. The official test set was used only after selection to measure the final accuracy of the chosen SVM. Keeping the HOG features and data splits fixed allowed the results to isolate the effect of changing the classifier and its C value.

5.2 Candidate and Final Results

All three MNIST candidates completed successfully. C = 0.1 produced the highest validation accuracy, 97.64%, and was selected for final testing. It reached 99.95% training accuracy, 97.74% test accuracy, and a training time of 338.97 seconds. Increasing C to 1 and 10 raised training accuracy to 100%, but validation accuracy decreased to 97.29% and 97.11%. Training time also increased to 772.64 and 914.51 seconds, respectively.

Table 5. MNIST linear-SVM candidates

MNIST C	Training	Validation	Training time
0.1	99.95%	97.64%	338.97 s
1.0	100.00%	97.29%	772.64 s
10.0	100.00%	97.11%	914.51 s

Source: project notebook outputs.

C = 0.1 was also selected for CIFAR-10 because it produced the highest validation accuracy, 51.75%. Its training accuracy was 58.77%, and the selected model reached 50.86% test accuracy. Training and validation accuracy were almost unchanged across C = 0.1, 1, and 10, indicating that the CIFAR-10 SVM was not sensitive to C within the tested range.

Table 6. CIFAR-10 linear-SVM candidates

CIFAR-10 C	Training	Validation	Training time
0.1	58.77%	51.75%	239.96 s
1.0	58.78%	51.67%	239.22 s
10.0	58.77%	51.66%	238.15 s

Source: project notebook outputs.

Dataset	Inherited HOG	Selected C	Training	Validation	Final test	Training time
MNIST	12 bins; 1,728 features	0.1	99.95%	97.64%	97.74%	338.97 s
CIFAR-10	6 bins; 1,176 features	0.1	58.77%	51.75%	50.86%	239.96 s

Table 7. Selected linear-SVM configurations and required evidence

Source: project notebook outputs.

5.3 Analysis

Dataset	HOG logistic test	HOG SVM test	Accuracy change	Logistic time	SVM time
MNIST	97.86%	97.74%	-0.12 points	7.08 s	338.97 s
CIFAR-10	50.84%	50.86%	+0.02 points	29.04 s	239.96 s

Table 8. Selected HOG classifier comparison

Source: project notebook outputs.

Changing the classifier produced almost no accuracy improvement. On MNIST, HOG with logistic regression reached 97.86% test accuracy, while HOG with the linear SVM reached 97.74%, making logistic regression 0.12 percentage points more accurate. On CIFAR-10, the SVM reached 50.86%, only 0.02 points above logistic regression at 50.84%. The SVM also required substantially more training time in these runs. These results

suggest that the two classifiers found similarly useful linear boundaries when given the same standardized HOG features.

Increasing C did not improve validation accuracy on either dataset. For MNIST, the larger C values produced perfect training accuracy, but validation accuracy decreased and training took considerably longer. For CIFAR-10, training and validation accuracy changed very little across the three C values. The gaps between training and validation accuracy indicate some overfitting, so penalizing margin violations more strongly did not improve performance on unseen images.

The much larger performance change occurred earlier, when the representation changed from raw pixels to HOG while the classifier remained logistic regression. Changing from logistic regression to a linear SVM while keeping HOG fixed made almost no difference. This supports the conclusion that feature representation mattered more than the choice between these two linear classifiers in these experiments.

6 Work Package 4 Fully Connected Neural Networks

6.1 Architecture and Training

The fully connected networks learned nonlinear combinations of flattened pixel values. ToTensor converted the images to tensors and scaled their pixel values from the 0-255 range to the 0-1 range, and the Flatten layer converted each image into a one-dimensional vector. The MNIST architecture was Flatten - Linear(784, 128) - ReLU - Linear(128, 10), and it was trained for 10 epochs with Adam, a learning rate of 0.001, batch size 128, and cross-entropy loss. The CIFAR-10 architecture was Flatten - Linear(3072, 512) - ReLU - Linear(512, 256) - ReLU - Linear(256, 10). For the final controlled CIFAR-10 comparison, the architecture, Adam optimizer, cross-entropy loss, batch size of 128, fixed data split, and random seed were held constant while models using learning rates of 0.01, 0.001, and 0.0001 were trained for 50 epochs.

Before that final comparison, several experiments attempted to improve CIFAR-10 validation accuracy. First, an additional hidden layer was added, but validation accuracy decreased, so the original architecture was restored. Next, batch sizes of 64 and 256 were compared with the standard batch size of 128 during exploratory 25-epoch runs; these results are summarized in Table 11. Finally, training was increased from 25 to 50 epochs with the original architecture and batch size 128. This last change produced the strongest final validation result when paired with the learning rate of 0.0001.

6.2 MNIST Results

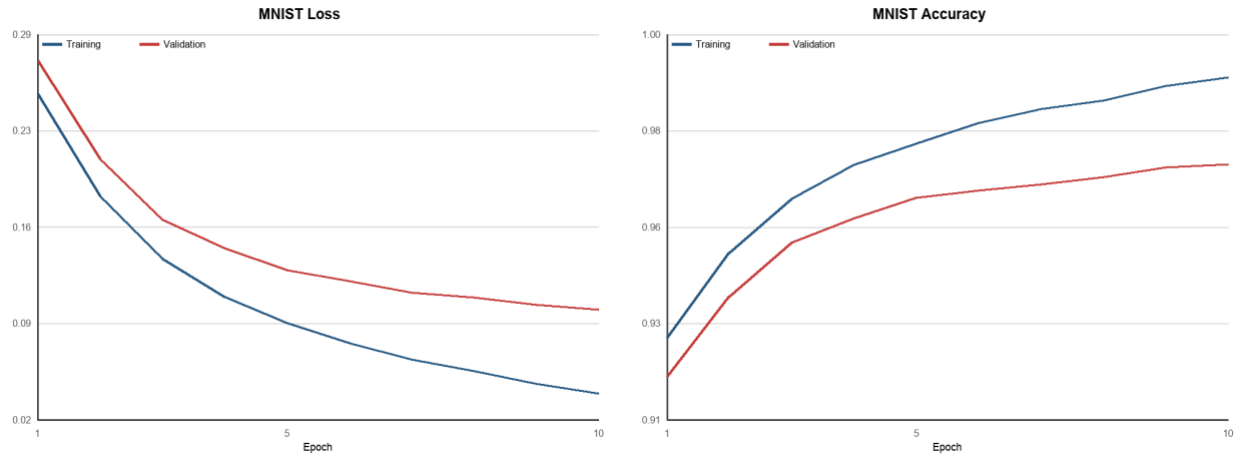


Figure 1. MNIST fully connected network training and validation curves

MNIST training accuracy increased from 93.04% after the first epoch to 99.12% after epoch 10. Validation accuracy increased from 92.13% to 97.09%, and the final test accuracy was 97.56%. Training required 53.74

seconds on the T4 GPU. Both losses decreased throughout training, so the network was still learning. The approximately two-percentage-point training-validation gap indicates mild overfitting, but the close validation and test accuracies show that the selected model generalized consistently.

6.3 CIFAR-10 Learning Rate Investigation

Learning rate	Final training	Final validation	Training time
0.01	37.81%	35.19%	276.80 s
0.001	75.65%	50.83%	293.93 s
0.0001	67.68%	53.64%	289.58 s

Table 9. CIFAR-10 learning-rate candidates after 50 epochs

Source: project notebook outputs.

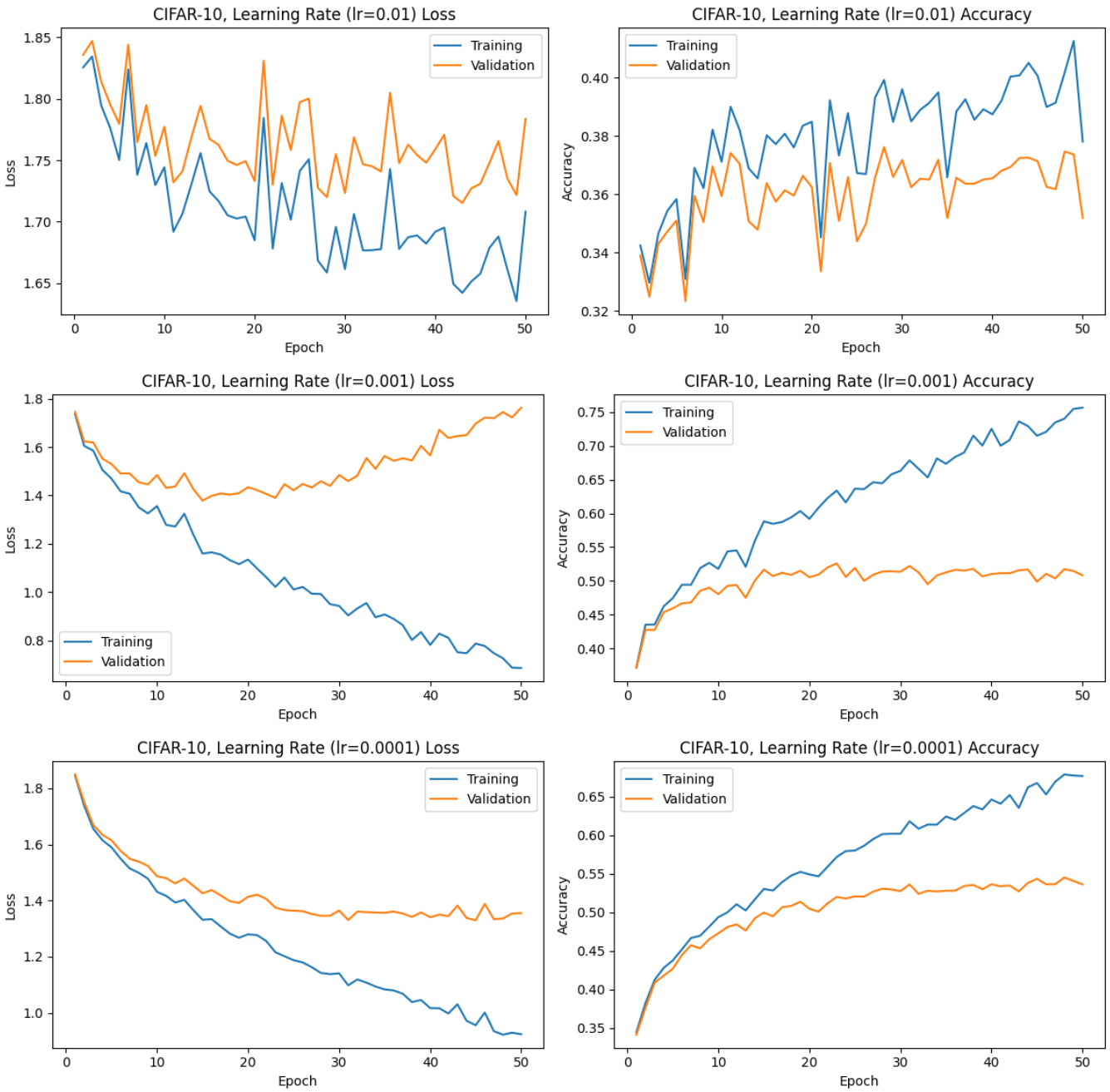


Figure 2. CIFAR-10 training and validation curves for all three learning rates

The learning rate of 0.01 remained unstable and ended at only 35.19% validation accuracy. The 0.001 model learned more quickly, but its training accuracy continued to rise while its validation accuracy leveled off and finished at 50.83%, showing overfitting. The 0.0001 model improved more gradually and finished with the highest validation accuracy, 53.64%. Because the selection rule used final validation accuracy, 0.0001 was selected for testing.

The selected 50-epoch CIFAR-10 network reached 53.96% test accuracy with a test loss of 1.3404. Its final training accuracy was 67.68%, creating a 14.04-percentage-point gap from validation accuracy. The curves therefore show that the lower learning rate produced the best final validation result, but the model still overfit as training continued.

Table 10. Selected CIFAR-10 fully connected network and required evidence

Dataset	Selected settings	Final training	Final validation	Final test	Training time
CIFAR-10	lr=0.0001; batch 128; 50 epochs	67.68%	53.64%	53.96%	289.58 s

Source: project notebook outputs.

6.4 Batch Size and Epoch Follow-Up

Table 11 lists exploratory follow-up runs using the same architecture, optimizer, loss, seed, and data split. Because the batch-256 run used a different learning rate, it does not provide a strictly controlled batch-size comparison. The final 50-epoch comparison fixed batch size at 128 and varied only learning rate.

Experiment	Batch	Epochs	Learning rate	Final validation	Test
25-epoch selection	128	25	0.0001	52.08%	52.17%
Batch-size trial	64	25	0.0001	50.68%	Not tested
Batch-size trial	256	25	0.001	51.28%	Not tested
50-epoch selection	128	50	0.0001	53.64%	53.96%

Table 11. CIFAR-10 batch-size and epoch follow-up experiments

Source: project notebook outputs.

Batch size 128 produced the highest validation result among the recorded follow-up runs, although the different learning rate used for the batch-256 run prevents a strict batch-size comparison. With 50 epochs, the learning rate of 0.0001 reached 53.64% final validation accuracy and 53.96% test accuracy. This improved the 25-epoch test result by 1.79 points, but the training-validation gap widened to 14.04 points and training time increased from 140.23 to 289.58 seconds, providing clearer evidence of overfitting.

6.5 Comparison with Classical Pipelines

ReLU hidden layers allow a fully connected network to learn nonlinear combinations of pixels, unlike logistic regression and the linear SVM. However, flattening still removes the image's explicit two-dimensional arrangement. The network must learn useful relationships among pixel positions without the spatial assumptions provided by HOG or a convolutional layer.

On MNIST, the network's 97.56% test accuracy was close to the HOG pipelines but did not exceed HOG logistic regression at 97.86%. Its training and validation curves remained fairly close, indicating only mild overfitting. On CIFAR-10, the selected 50-epoch network produced the strongest result among the first four pipelines, 53.96%. This exceeded HOG logistic regression by 3.12 percentage points and HOG with the linear SVM by 3.10 points. However, CIFAR-10 training accuracy reached 67.68% while validation accuracy ended at 53.64%. This widening gap is evidence of overfitting: the model continued to fit the training images more closely after its performance on unseen validation images had begun to level off.

7 Work Package 5 Convolutional Neural Networks

7.1 Architecture and Training

The convolutional networks processed the ToTensor-scaled images in their original spatial form rather than flattening them before feature extraction. Both networks used two convolutional blocks. Each block contained a 3 by 3 convolution with padding of 1, a ReLU activation, and 2 by 2 max pooling. The first convolution

produced 32 feature maps and the second produced 64. After two pooling operations, the feature maps were flattened and passed through a 128-node fully connected layer, ReLU, and a ten-node output layer.

MNIST entered the network as tensors with one channel and 28 by 28 pixels. Pooling reduced the spatial size to 7 by 7, producing 64 x 7 x 7 = 3,136 inputs to the first fully connected layer. CIFAR-10 entered as tensors with three channels and 32 by 32 pixels and was reduced to 64 x 8 x 8 = 4,096 values. Both models used Adam, cross-entropy loss, a learning rate of 0.001, batch size 128, and seed 1234. MNIST was trained for 10 epochs and CIFAR-10 for 50 epochs on a Colab T4 GPU.

7.2 MNIST Results

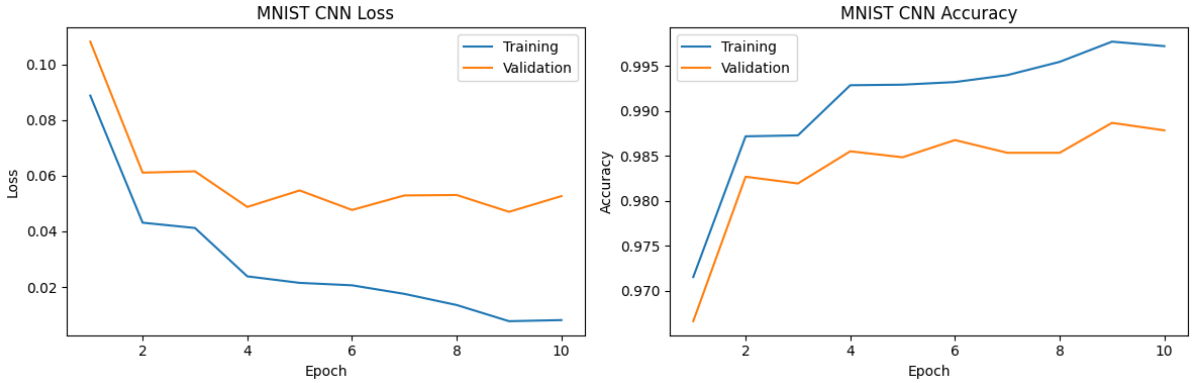


Figure 3. MNIST CNN training and validation curves

The MNIST CNN reached 99.72% final training accuracy, 98.78% final validation accuracy, and 99.02% test accuracy in 62.66 seconds. Validation accuracy peaked at 98.87% in epoch 9 and remained close to training accuracy, while validation loss stayed low. This indicates only mild overfitting. Compared with the fully connected network, the CNN improved test accuracy by 1.46 percentage points.

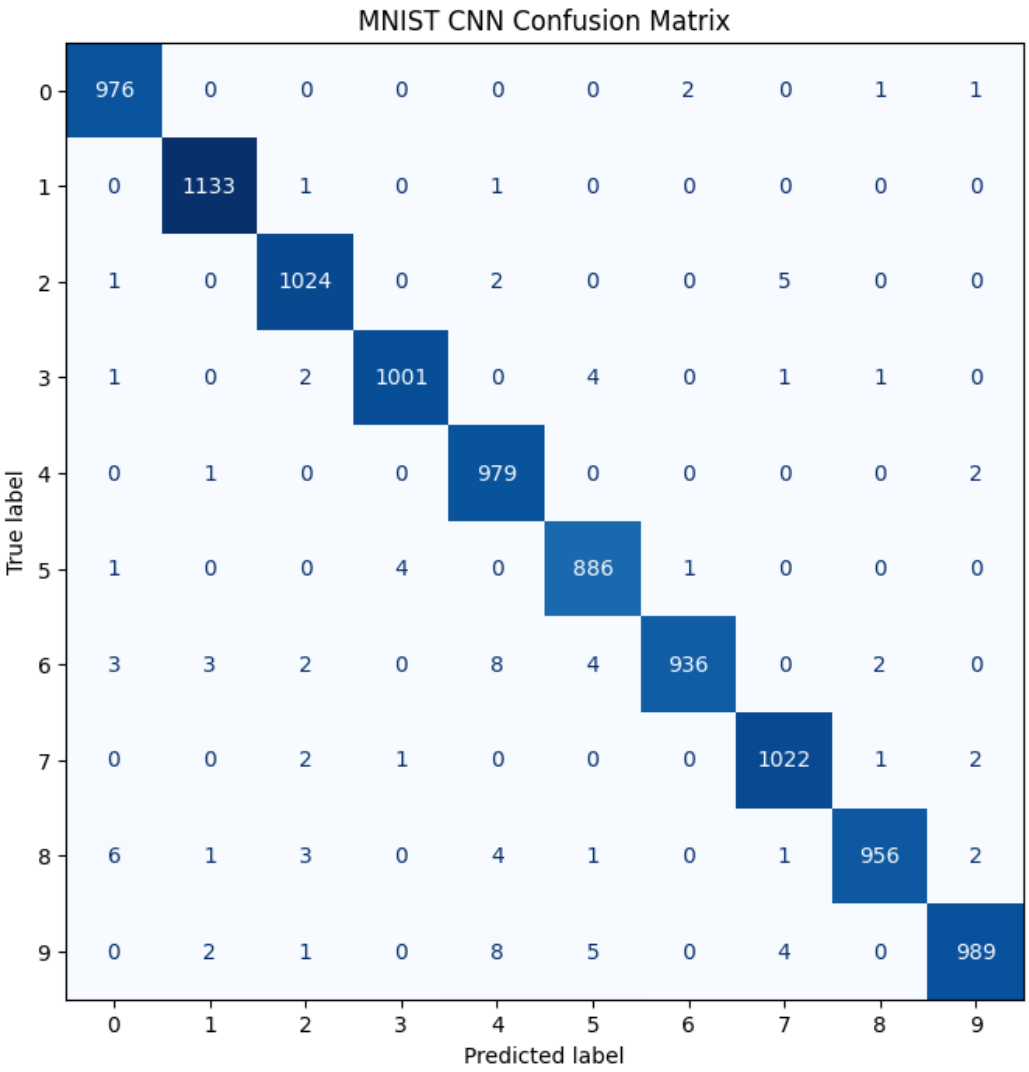


Figure 4. MNIST CNN test confusion matrix

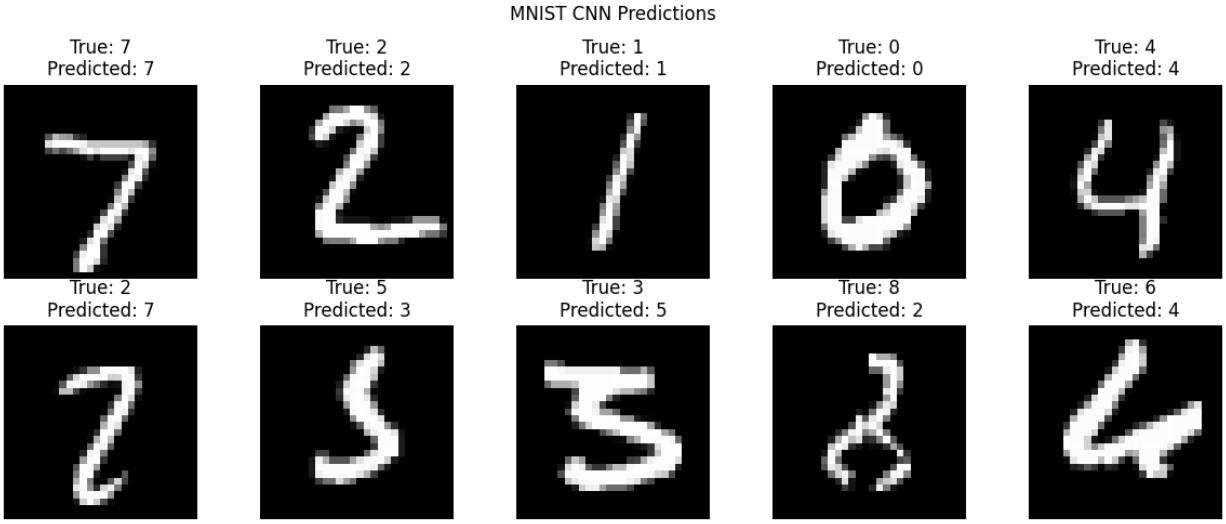


Figure 5. Five correct and five incorrect MNIST CNN predictions

The confusion matrix is strongly concentrated on the diagonal, confirming that errors were rare for every class. The most frequent individual confusions included 6 predicted as 4 and 9 predicted as 4, with eight cases each. The displayed mistakes are understandable because several digits are written in unusual or ambiguous ways: a curved 2 resembles a 7, a 5 resembles a 3, and an open 6 resembles a 4.

7.3 CIFAR-10 Results

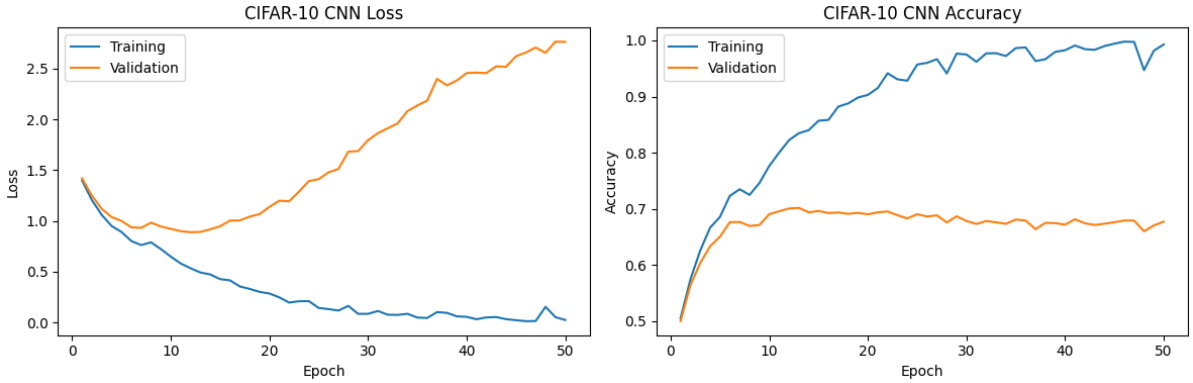


Figure 6. CIFAR-10 CNN training and validation curves

The CIFAR-10 CNN reached 99.28% final training accuracy, 67.71% final validation accuracy, and 67.28% test accuracy in 294.85 seconds. Validation accuracy peaked at 70.16% in epoch 13, then remained near 67% while training accuracy continued toward 100%. At the same time, training loss approached zero while validation loss rose to 2.7621. These curves provide strong evidence of overfitting after the early portion of training.

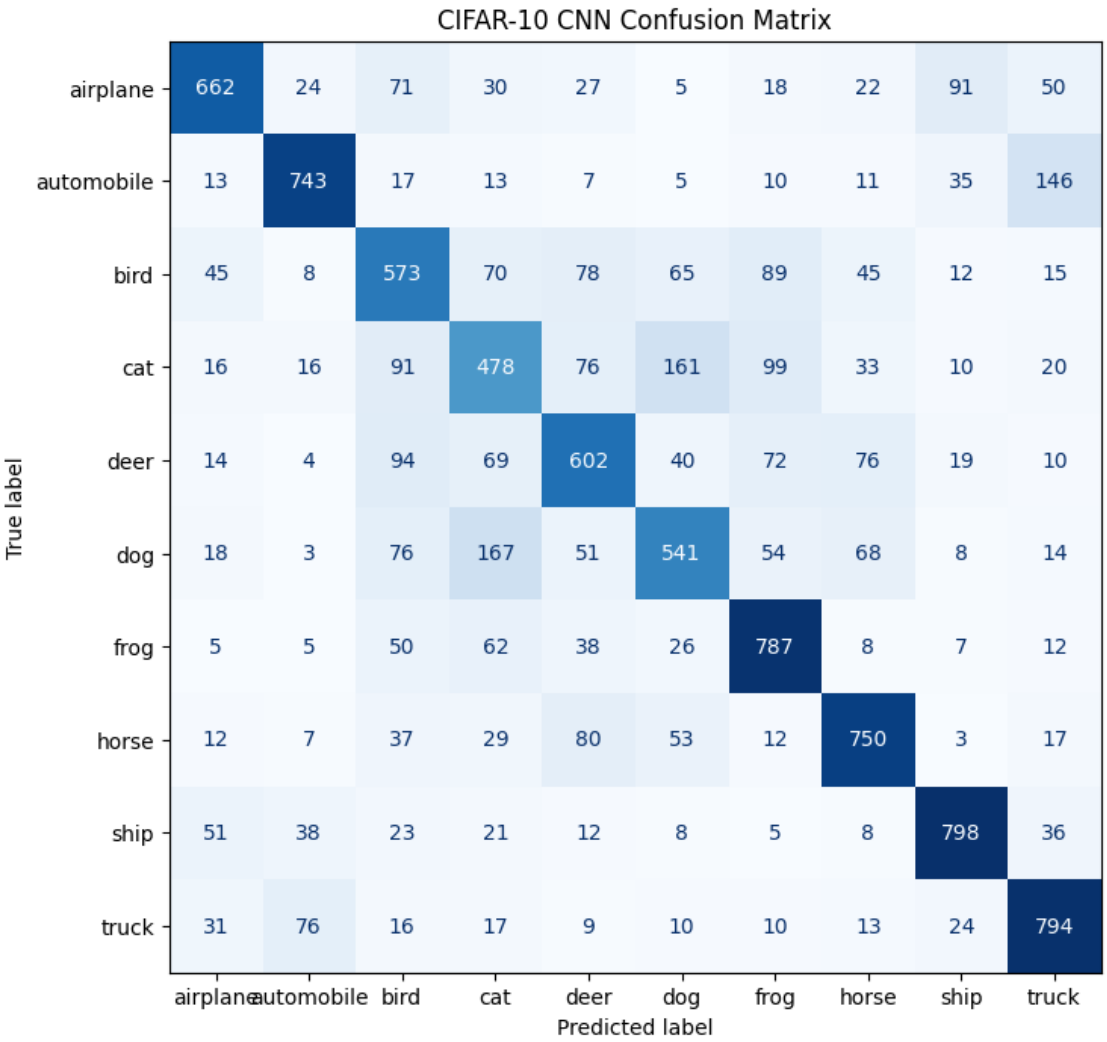


Figure 7. CIFAR-10 CNN test confusion matrix

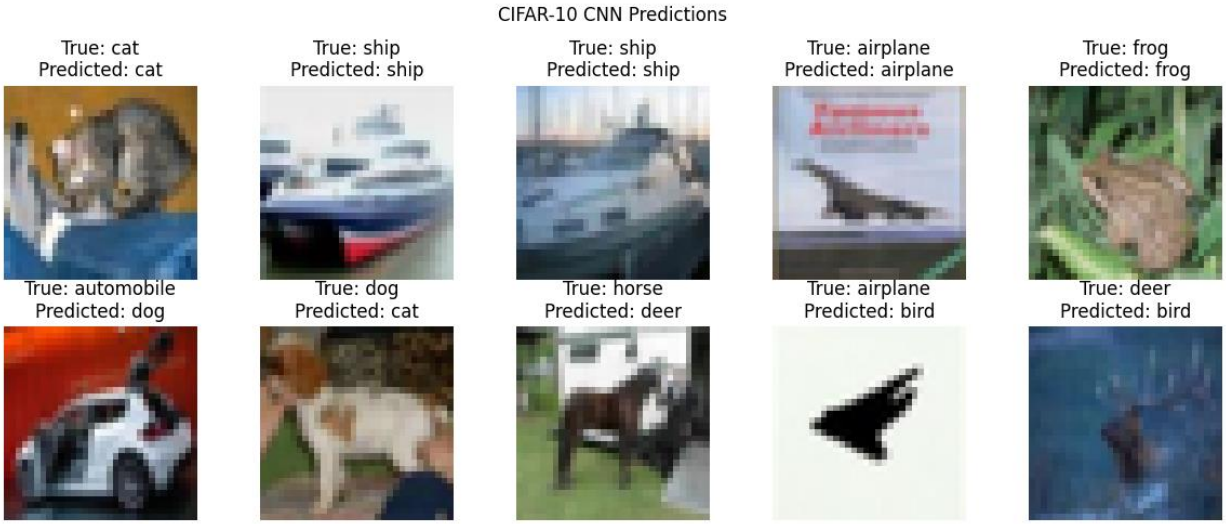


Figure 8. Five correct and five incorrect CIFAR-10 CNN predictions

The most frequent CIFAR-10 confusions occurred between visually related classes. Cats and dogs were confused in both directions, with 161 cats predicted as dogs and 167 dogs predicted as cats. The network also predicted 146 automobiles as trucks and 91 airplanes as ships. These mistakes are reasonable for 32 by 32 images because related animals share shapes and textures, while vehicles can have similar outlines or backgrounds. The example errors also show that small objects, cluttered scenes, and limited detail can make the true class difficult even for a person to identify quickly.

A fully connected network assigns separate weights to fixed pixel positions after flattening. A CNN instead slides shared filters across neighboring pixels, allowing the same learned feature to be detected in different locations. Pooling reduces spatial dimensions and makes the representation less sensitive to small shifts. This spatial approach produced a modest improvement on the centered and consistent MNIST digits, but it improved CIFAR-10 test accuracy by 13.32 percentage points. CIFAR-10 benefited more because recognizing its objects depends heavily on local edges, textures, and parts that may appear in different positions.

8 Work Package 6 Cross-Pipeline Synthesis

8.1 Consolidated Results

Dataset	Representation	Classifier	Selected settings	Validation	Test	Time
MNIST	Raw pixels	Logistic regression	784 features; C=1	91.36%	92.30%	10.17 s
MNIST	HOG	Logistic regression	12 bins; 1,728 features	97.71%	97.86%	7.08 s
MNIST	HOG	Linear SVM	12 bins; C=0.1	97.64%	97.74%	338.97 s
MNIST	Flattened pixels	Fully connected NN	128 hidden; lr=.001; 10 ep.	97.09%	97.56%	53.74 s
MNIST	Spatial tensor	CNN	32-64 channels; lr=.001; 10 ep.	98.78%	99.02%	62.66 s
CIFAR-10	Raw pixels	Logistic regression	3,072 features; C=1	36.24%	36.55%	304.80 s
CIFAR-10	HOG grayscale	Logistic regression	6 bins; 1,176 features	51.57%	50.84%	29.04 s
CIFAR-10	HOG grayscale	Linear SVM	6 bins; C=0.1	51.75%	50.86%	239.96 s
CIFAR-10	Flattened pixels	Fully connected NN	512-256; lr=.0001; 50 ep.	53.64%	53.96%	289.58 s
CIFAR-10	Spatial RGB tensor	CNN	32-64 channels; lr=.001; 50 ep.	67.71%	67.28%	294.85 s

Table 12. Consolidated final results for all ten pipelines

Source: project notebook outputs. HOG times exclude feature extraction; neural-network times exclude the evaluation passes performed after each epoch.

8.2 Selected HOG SVM Confusion Matrices

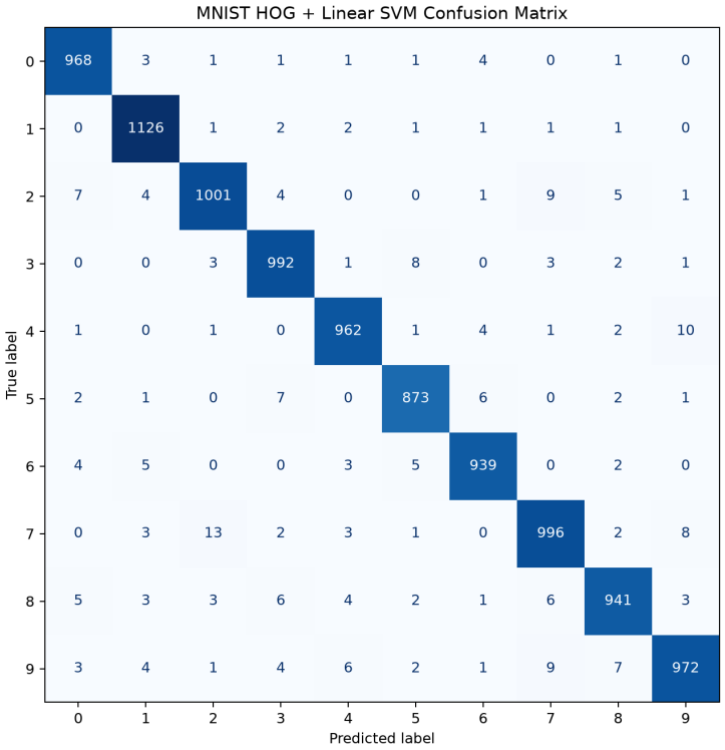


Figure 9. Selected MNIST HOG and linear SVM test confusion matrix

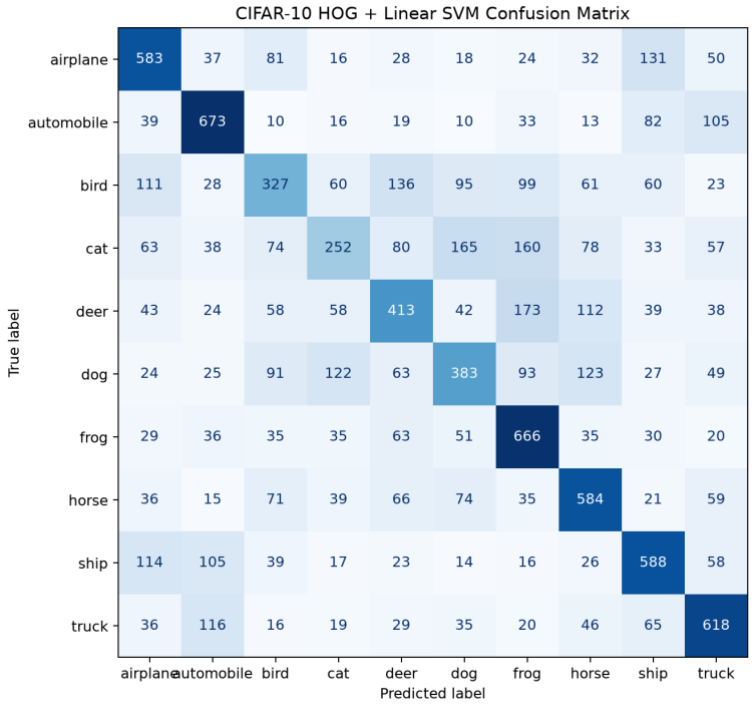


Figure 10. Selected CIFAR-10 HOG and linear SVM test confusion matrix

The selected HOG-SVM confusion matrices reinforce the accuracy results. MNIST classifications remain concentrated on the diagonal because digit outlines are well described by local edge directions. CIFAR-10 errors are spread across more classes because the selected grayscale HOG representation discarded color and described local edge structure without learning higher-level object parts. The CIFAR-10 CNN matrix is more strongly diagonal than the CIFAR-10 HOG-SVM matrix because the CNN learns spatial features directly from the RGB images.

8.3 Overall Findings

All methods performed better on MNIST, where test accuracy ranged from 92.30% to 99.02%, than on CIFAR-10, where it ranged from 36.55% to 67.28%. Centered grayscale digits make fixed locations informative, while CIFAR-10's changing backgrounds, viewpoints, positions, colors, and within-class appearances demand stronger representations of local and spatial structure.

The largest MNIST transition was from raw pixels to HOG, a gain of 5.56 percentage points. HOG also produced the largest individual CIFAR-10 gain, 14.29 points, closely followed by the 13.32-point gain from the fully connected network to the CNN. Both changes improved how image structure was represented. Replacing logistic regression with a linear SVM while keeping HOG fixed changed accuracy by only -0.12 points on MNIST and +0.02 points on CIFAR-10, so representation mattered much more than the choice between the two linear classifiers.

The most surprising result was how little the classifier change mattered once HOG was fixed. HOG logistic regression even slightly exceeded the fully connected network on MNIST, showing that a simple model can remain competitive when its representation matches the data. The CNN led both datasets, with a larger CIFAR-10 advantage because shared filters and pooling preserve local relationships and recognize features in different positions.

9 Limitations

- The experiments used one fixed split and random seed. Repeating the experiments with additional splits or seeds would provide a stronger estimate of result variability.
- Only the required small hyperparameter sets were investigated.
- The final CIFAR-10 CNN was evaluated after epoch 50 even though validation accuracy peaked earlier. Its final curves should therefore be interpreted as both a performance result and evidence of overfitting.

10 Conclusion

The project demonstrates that classification performance depends strongly on how an image is represented. Raw pixels were sufficient for a useful MNIST baseline but weak for CIFAR-10. HOG improved both datasets by converting pixels into local edge and shape descriptions, while changing from logistic regression to a linear SVM had almost no effect when those features remained fixed. Fully connected networks added nonlinear learning but still discarded explicit spatial layout. The CNN achieved the highest accuracy by learning local spatial features directly from the original tensors, reaching 99.02% on MNIST and 67.28% on CIFAR-10. The growing CIFAR-10 training-validation gap also shows that greater model capacity must be balanced with methods that control overfitting.

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Generative AI was used as a study and drafting aid to discuss course concepts, review code structure and outputs, and help organize an initial version of this report. I remained responsible for implementing and running the experiments, checking the recorded results, understanding the submitted code, and revising the final report in accordance with course policy.